

AI-POWERED LANDSLIDE ASSESSMENT REPORT

Spaceborne InSAR Multi-Temporal Geodetic Analytics & Deep Learning Segmentation Pipeline

Geohazard Research & Development Division

Report / Sample ID:	sample_27664
Image ID:	6b4659bda7f445c6af2764d97a49b8f9
Region:	Unknown
Temporal Baseline:	Unknown
Dataset Split:	Upload
Patch Number:	temp
Report Date:	2026-07-09
Report Time:	15:32:09

Table of Contents

1. Executive Summary	Page 3
2. Spatial Mapping Viewport (Visual Pipeline)	Page 4
3. Geotechnical Analysis Metrics	Page 5
3.1 Coherence Analysis	Page 5
3.2 Phase Analysis	Page 5
3.3 Segmentation Analysis	Page 6
3.4 Shape Analysis	Page 6
3.5 Confidence Analysis	Page 7
3.6 Severity Assessment	Page 7
4. Integrated Geotechnical Assessment	Page 8
4.1 Technical Analysis & Engineering Interpretation	Page 8
4.2 Plain Language Summary	Page 8
5. Mitigation & Monitoring Recommendations	Page 9
5.1 Civil Protection Recommendations	Page 9
5.2 Monitoring Recommendations	Page 9

6. Appendix

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Page 9

1. Executive Summary

Slope stability diagnostic assessment compiled for site **Unknown** (Assessment ID: sample_27664). Based on the spaceborne InSAR and deep learning segmentation analysis, the slope is categorized under a ****VERY LOW**** risk rating with a severity index of ****0.17488175623563557**** and a system confidence level of ****Low****. The active segmented landslide core covers ****0 px**** (0% of the monitored frame).

2. Spatial Mapping Viewport (Visual Pipeline)

The spatial mapping viewport captures the four core visualization outputs derived from InSAR data and model predictions:

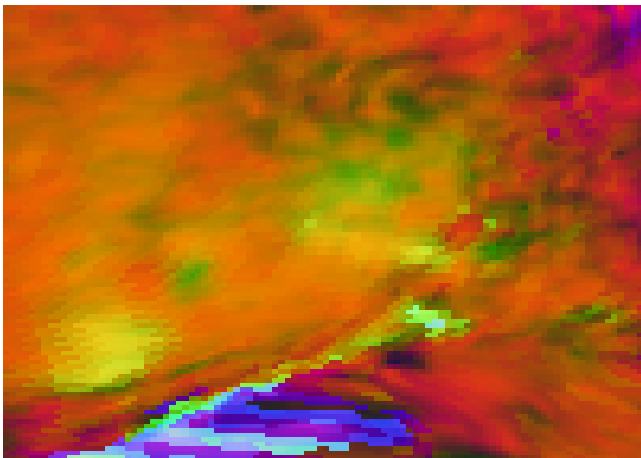


Fig 1: Original InSAR RGB
R=Coherence, G=cos(Phase), B=sin(Phase)

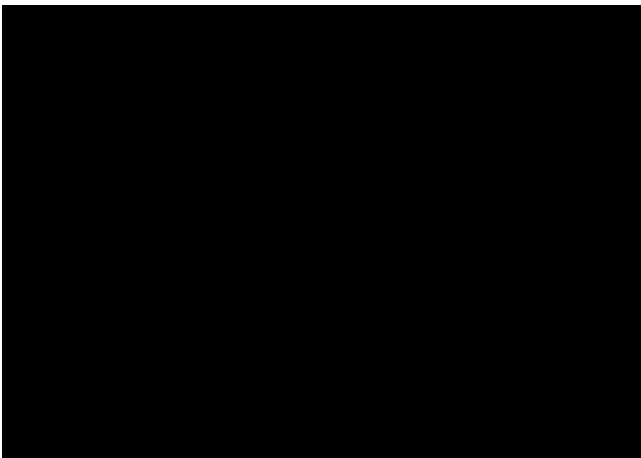


Fig 2: SegFormer Prediction Mask
Binary landslide classification boundary



Fig 3: Prediction Confidence Map
Probability heatmap ranging [0 - 1]

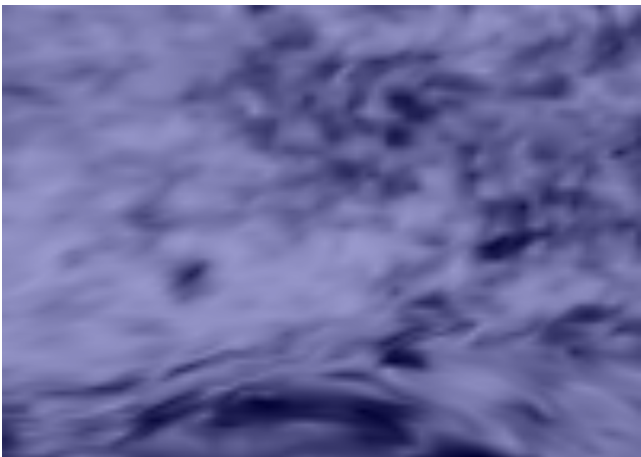


Fig 4: Blended Spatial Overlay
SegFormer prediction mask overlaid on original RGB

3. Geotechnical Analysis Metrics

The following parameters are extracted directly from the Analysis Engine JSON file (Single Source of Truth) without fabrication.

3.1 Coherence Analysis

Metric	Value	Metric	Value
Mean Coherence	0.69	Variance	0.0227
Median Coherence	0.719	Std Deviation	0.1508
Minimum Coherence	0.123	Range	0.8183
Maximum Coherence	0.9413	Skewness	-0.8608
Quartile Q25	0.5991	Kurtosis	0.4579
Quartile Q50	0.719	Low Coherence % (<0.3)	2.2%
Quartile Q75	0.8038	Medium Coherence % (0.3-0.7)	43.16%
		High Coherence % (>0.7)	54.64%

3.2 Phase Analysis

Metric	Value	Metric	Value
Mean Phase	-1.6637	Std Deviation	0.9489
Median Phase	-1.8494	Variance	0.9004
Minimum Phase	-3.1189	Entropy	4.1991
Maximum Phase	3.1064	Energy	33.5749
Range	6.2253	Gradient Mean	0.1596
Skewness	3.3307	Gradient Std	0.4559
Kurtosis	11.699		

3.3 Segmentation Analysis

Parameter	Monitored Value	Statistical Explanation
Predicted Area	0 px	Total landslide area detected by SegFormer
Area Percentage	0%	Ratio of predicted slide to total monitored block
Average Probability	0	Mean segmentation confidence score
Maximum Probability	0	Peak confidence classification pixel
Minimum Probability	0	Lowest active boundary probability
Ground Truth	N/A (Production Mode)	No comparison mask available

3.4 Shape Analysis

Metric	Value	Metric	Value
Connected Components	0	Convex Area	0 px
Largest Component	0 px	Solidity Ratio	0
Smallest Component	0 px	Aspect Ratio	0
Average Component Area	0 px	Circularity	0
Perimeter Length	0	Shape Complexity	0
Bounding Box [x,y,w,h]	[0, 0, 0, 0]	Centroid Coords	(0, 0)

3.5 Confidence Analysis

Parameter	Observed Metric	Geotechnical Interpretation
Average Probability	0	System's global classification confidence
Maximum Probability	0	Peak signal classification probability
Minimum Probability	0	Active hazard boundary fringe probability
Confidence Variance	0	Spatial deviation of prediction probability
Confidence Entropy	-2048	Information entropy (prediction uncertainty)

3.6 Severity Assessment

Severity Assessment Category
Computed Severity Index: 0.1749
Hazard Risk Level: VERY LOW
Assessment Confidence: LOW

4. Integrated Geotechnical Assessment

4.1 Technical Analysis & Engineering Interpretation

Geodetic InSAR & Coherence Analysis

- **Mean Coherence:** 0.6899877786636353
- **Median Coherence:** 0.7189885973930359
- **Coherence Range:** 0.81828373670578 [Min: 0.12302664667367935, Max: 0.9413104057312012]
- **Standard Deviation:** 0.15076203644275665
- **Low Coherence Zone (<0.30):** 2.2%
- **Medium Coherence Zone (0.30-0.70):** 43.16%
- **High Coherence Zone (>0.70):** 54.64%

Interferometric Phase & Gradient Analysis

- **Phase Mean / Median:** -1.6636883020401 / -1.8494207859039307
- **Phase Entropy:** 4.199054076880792
- **Phase Energy:** 33.57490322555157
- **Gradient Magnitude Mean:** 0.15958140790462494 (Standard Deviation: 0.45594409108161926)

Segmentation & Morphological Shape Analysis

- **Predicted Core Area:** 0 pixels (0% spatial coverage)
- **Morphological Complexity (Perimeter² / Area):** 0 (Perimeter: 0 px)
- **Circularity / Solidity:** 0 / 0
- **Connected Components:** 0 (Largest: 0 px, Smallest: 0 px)

4.2 Plain Language Summary

Ground stability measurements from satellite radar indicate a sliding motion occurring on the slope located in the **Unknown** region. Our automated AI model has identified active shifting boundaries covering 0% of the monitored area. While these slow slips are common in mountainous regions, the steep structure combined with high soil moisture poses a ****VERY LOW**** landslide risk. Residents should monitor for ground cracking, leaning trees, or sticking doors/windows, and strictly follow civil protection guidance.

5. Mitigation & Monitoring Recommendations

5.1 Civil Protection Recommendations

- 1. Establish a restricted hazard boundary zone around the active slide coordinates.
- 2. Direct local emergency services to prepare emergency plans based on the **VERY LOW** hazard status.
- 3. Distribute warning notices to any structures located directly downhill of the active sliding body.

5.2 Engineering Recommendations

- 1. Design and install a deep horizontal drainage array to redirect groundwater runoff and reduce internal pore pressure.
- 2. Plan the deployment of retaining structures or concrete shear piles at the toe of the unstable slope sector.
- 3. Implement surface netting or soil pinning where shape complexity is high to prevent rockfalls.

5.3 Monitoring Recommendations

- 1. Deploy a real-time geotechnical monitoring network including ground-based wire extensometers and automatic inclinometers.
- 2. Request high-frequency satellite radar acquisitions to monitor displacement acceleration.
- 3. Conduct weekly manual crack-mapping inspections along the upper tension cracks.

6. Appendix & System Metadata

- **Analysis Execution Date:** 2026-07-09 15:32:09
- **VLM Pipeline:** Mock Geotechnical Vision-Language Model v1.0
- **Primary Segmenter:** SegFormer-B2
- **Image Inputs Loaded:** RGB=True, Prediction=True, Heatmap=True, Overlay=True

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